

EXPERIMENT 2

Case Study: Software Project Plan for an EV Battery Health and Transparency System

Aim

To prepare a software project plan for developing an EV Battery Health and Transparency System that monitors battery degradation, estimates driving range, analyzes charging behavior, and provides users with transparent information about battery condition.

Objectives

- Define the scope of the proposed EV battery monitoring application.
- Identify the major features and modules of the system.
- Prepare a development timeline for the project.
- Define the roles and responsibilities of the project team.
- Identify major project deliverables.
- Establish a systematic plan for developing the application.

Project Overview

The proposed system is a software application designed to provide EV users with detailed information about their vehicle battery. Instead of displaying only battery percentage and estimated range, the system provides Battery State of Health (SOH), State of Charge (SOC), battery degradation, estimated driving range, charging cycle count, battery temperature, fast-charging history, remaining useful battery life, charging recommendations, and battery health alerts.

Problem Statement

EV users generally receive limited information about the long-term condition of their battery. A dashboard may show that the battery is at 60%, but the user may not know whether the battery has 95% or 75% of its original capacity. This uncertainty can create range anxiety and make it difficult for users to understand whether charging and driving habits are affecting battery life. The proposed application aims to provide clear and understandable battery health information.

Project Scope

In Scope

- User registration and login
- Battery health dashboard
- SOH and SOC monitoring
- Range estimation
- Battery degradation tracking
- Charging history
- Temperature monitoring
- Charging recommendations
- Battery health alerts
- Graphs and reports

Out of Scope

- Physical battery repair
- Direct control of the vehicle battery
- Battery manufacturing
- Hardware modification

- Battery-cell replacement
- Direct control of charging equipment

Major System Modules

Module	Description
User Module	Registration, login, and user profile
Battery Health Module	Displays SOH and battery capacity
Range Module	Estimates remaining driving range
Degradation Module	Tracks battery health over time
Charging Module	Records charging behavior and cycles
Temperature Module	Monitors battery temperature
Analytics Module	Provides graphs and reports
Alert Module	Provides battery and maintenance warnings

Proposed Technology Stack

Component	Proposed Technology
Frontend	React / HTML / CSS / JavaScript
Backend	Python / Node.js
Database	MySQL / PostgreSQL
Data Analysis	Python
Visualization	Matplotlib / Chart.js
API	REST API
Cloud	AWS / Google Cloud
Version Control	GitHub

Project Timeline

Phase	Activity	Duration
1	Requirement Analysis	Week 1
2	System Design	Week 2
3	Database Development	Week 3
4	Frontend Development	Week 4–5
5	Backend Development	Week 5–6
6	Battery Analytics	Week 7
7	Integration	Week 8
8	Testing	Week 9
9	Deployment	Week 10

Phase	Activity	Duration
10	Documentation	Week 11

Team Roles

Team Role	Responsibilities
Project Manager	Planning, coordination, and monitoring
UI/UX Developer	Dashboard and user interface
Frontend Developer	Application interface development
Backend Developer	APIs and server-side logic
Data Analyst	Battery degradation and range analysis
Database Developer	Database design and management
Tester	Test cases and software testing
Documentation Lead	Reports and project documentation

Major Deliverables

Phase	Deliverable
Requirement Analysis	Requirement Specification
Design	System Architecture and UI Design
Development	Working Application
Data Analysis	Battery Degradation Model
Testing	Test Report
Deployment	Working Prototype
Documentation	Final Project Report

Expected Outcome

Example dashboard: Battery Health: 86% | Capacity Loss: 14% | Estimated Range: 390 km | Charge Cycles: 800 | Battery Temperature: 31°C | Health Status: Good. This provides substantially greater transparency than simply displaying battery percentage.

Conclusion

The project plan provides a structured approach for developing an EV Battery Health and Transparency System. By clearly defining scope, timeline, team responsibilities, modules, technologies, and deliverables, the project can be developed systematically. The proposed system aims to provide meaningful battery information to EV users, helping them understand degradation, make better charging decisions, and reduce range anxiety.